SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

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Code of Conduct:

I _____B.Vinayani192472179_______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and

must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

ASSESSMENT TOOL 1

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and
Data Link layers ensure reliable communication.

Ans **File Transfer (150 MB)**

- File size = 150 MB = 150 × 10^6 = **150,000,000 bytes**
- Segment size = 1500 bytes

Number of segments:

- 150,000,000 / 1500 = **100,000 segments**

Number of frames:

- One segment per frame = **100,000 frames**

Data Link overhead:

- 40 bytes/frame × 100,000 = **4,000,000 bytes (4 MB)**

Total data transmitted:

- 150,000,000 + 4,000,000 = **154,000,000 bytes**
- = 1,232,000,000 bits

Transmission time:

- Bandwidth = 50 Mbps
- Time = 1,232,000,000 / 50,000,000 = **24.64 s**

Reliability:

- Transport Layer: segmentation, acknowledgments, error recovery.
- Data Link Layer: framing, CRC/error detection, MAC addressing.

2. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Ans **Sliding Window (48 Frames)**

- Window size = 6
- Total frames = 48

Transmission windows:

- 48 / 6 = **8 windows**

Retransmissions:

- 1 frame lost per window
- Total = **8 retransmissions**

Flow Control:

- Prevents sender from overwhelming the receiver.
- Improves throughput and reduces congestion.

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Ans **IP Fragmentation**

- Packet size = 12,000 bytes
- MTU = 1500 bytes
- IP header = 20 bytes

Data per fragment:

- `1500 − 20 = 1480 bytes`

Fragments required:

- 12,000 / 1480 = 8.1

- Therefore = 9 fragments

Header overhead:

- 9 × 20 = 180 bytes

Network Layer Role:

- Performs fragmentation and reassembly.
- Uses identification, fragment offset, and flags for correct packet reconstruction.

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Ans **Sliding Window**

- Windows = 48/6 = **8**
- Retransmissions = **8**

Flow Control:

- Maintains efficient communication by matching sender speed to receiver capacity.

5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Ans **Session Layer Synchronization**

- File size = 600 MB
- Checkpoint every 60 MB
- Failure after 390 MB

Checkpoints before failure:

- 60, 120, 180, 240, 300, 360
- Total = **6 checkpoints**

Data to retransmit:

- $390 - 360 =$ **30 MB**

Importance:

- Allows transmission to resume from the last checkpoint.
- Saves time and bandwidth.

6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Ans **Presentation Layer Compression**

- Original size = 900 MB
- Compression = 40%

Compressed size:

- 900 × 0.60 = **540 MB**

Encrypted size:

- 540 × 1.05 = **567 MB**

Percentage reduction:

- $((900 - 567)/900) \times 100$
- = **37%**

Presentation Layer Benefits:

- Compression improves efficiency.
- Encryption provides confidentiality and security.

7.An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Ans **FTP Data Transfer**

- Data = 3 GB = 3000 MB
- Request size = 3 MB

Requests generated:

- 3000/3 = **1000 requests**

Transmission time:

- 3 GB = 24,000 Mb
- Time = 24,000/120
- = **200 seconds**

Application Layer Services:

- Provides user interfaces and protocols (FTP, HTTP, SMTP).
- Ensures reliable file transfer and resource access.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Ans **End-to-End Delay**

Component	**Delay**

Application	4 ms
Transport	3 ms
Network	5 ms
Data Link	2 ms
Physical	1 ms
Propagation	18 ms
Transmission	12 ms

Total Delay:

- 4 + 3 + 5 + 2 + 1 + 18 + 12
- = **45 ms**

Layer Contributions:

- Application: user services.
- Transport: reliability and segmentation.
- Network: routing.
- Data Link: framing and error checking.
- Physical: bit transmission.

9.A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Ans **Frame Efficiency**

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Overhead/frame = 80 bytes

Useful data/frame:

- $1600 - 80 =$ **1520 bytes**

Frames required:

- $80{,}000{,}000 / 1520 \approx$ **52,632 frames**

Total overhead:

- 52,632 × 80 = **4,210,560 bytes (~4.21 MB)**

Efficiency:

- (1520/1600) × 100
- = **95%**

Effect of Overhead:

- Increases bandwidth usage.
- Reduces effective throughput.
- Necessary for addressing, synchronization, and error detection.

10.A hospital transfers a patient's medical record of 240 MB through an IoT- enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Ans **Healthcare Network**

- Original size = 240 MB
- Compression = 30%

Compressed size:

- 240 × 0.70 = **168 MB**
- = 168,000,000 bytes

Segments:

- Segment size = 1400 bytes
- 168,000,000 / 1400 = **120,000 segments**

Protocol overhead:

- 60 bytes/frame × 120,000
- = **7,200,000 bytes (7.2 MB)**

Final data transmitted:

- 168,000,000 + 7,200,000
- = **175,200,000 bytes**
- = **175.2 MB**

OSI Layer Cooperation:

- Presentation: compression and encryption.
- Transport: segmentation and reliability.
- Data Link: framing and error detection.
- Physical: transmits bits.
- Together they ensure secure, reliable, and efficient communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand